

V4.1 BOM 核对清单（按位号顺序）

共 177 个位号，其中 167 个需要贴片。贴完一个在左侧方框打勾。坐标以 PCB 左上角为原点，单位 mm。

板上丝印印错了 —— 别信板上的字，认坐标

有极性/方向 —— 贴反不工作

不贴

贴	位号	值 / 型号	封装	位置 (x, y)	阶段	备注
	C1	10uF	C_0805_2012Metric	70.8, 96.5	A 电源	
	C2	1uF	C_0603_1608Metric	85.9, 99.1	A 电源	
	C3	1uF	C_0603_1608Metric	70.8, 98.9	A 电源	
	C4	1uF	C_0603_1608Metric	78.7, 97.8	A 电源	
	C5	1uF	C_0402_1005Metric	85.9, 96.7	A 电源	
	C6	1uF	C_0603_1608Metric	92.1, 97.9	A 电源	板上印着 C18
	C10	100nF	C_0603_1608Metric	61.0, 92.6	C 传感器 + GNSS	
	C11	100nF	C_0603_1608Metric	64.6, 87.2	C 传感器 + GNSS	
	C12	100nF	C_0603_1608Metric	56.5, 87.7	C 传感器 + GNSS	
	C13	100nF	C_0603_1608Metric	59.1, 92.6	C 传感器 + GNSS	
	C14	100nF	C_0603_1608Metric	64.6, 89.9	C 传感器 + GNSS	
	C15	4.7uF	C_0603_1608Metric	59.9, 99.3	C 传感器 + GNSS	
	C16	100nF	C_0603_1608Metric	64.6, 92.5	C 传感器 + GNSS	
	C17	100nF	C_0603_1608Metric	71.4, 81.0	C 传感器 + GNSS	
	C18	10uF	C_0805_2012Metric	74.4, 80.8	C 传感器 + GNSS	板上印着 C6
	C19	33pF	C_0603_1608Metric	88.4, 69.1	B MCU + Flash	
	C20	33pF	C_0603_1608Metric	80.0, 69.1	B MCU + Flash	
	C22	100nF	C_0603_1608Metric	94.5, 79.5	B MCU + Flash	
	C23	100nF	C_0603_1608Metric	94.5, 74.9	B MCU + Flash	
	C24	100nF	C_0603_1608Metric	80.0, 78.4	B MCU + Flash	
	C25	100nF	C_0603_1608Metric	82.8, 85.1	B MCU + Flash	板上印着 R33，尺寸相同会真贴错
	C26	100nF	C_0603_1608Metric	80.0, 75.8	B MCU + Flash	
	C27	100nF	C_0603_1608Metric	93.2, 81.8	B MCU + Flash	
	C28	1uF	C_0603_1608Metric	81.3, 82.8	B MCU + Flash	
	C29	1uF	C_0603_1608Metric	80.0, 81.0	B MCU + Flash	
	C30	100pF	C_0402_1005Metric	102.8, 67.5	E 1090 接收链	板上印着 L14，尺寸相同会真贴错
	C31	12pF	C_0402_1005Metric	111.3, 62.2	E 1090 接收链	
	C32	12pF	C_0402_1005Metric	114.3, 62.2	E 1090 接收链	
	C33	12pF	C_0402_1005Metric	120.4, 63.2	E 1090 接收链	
	C34	3pF	C_0402_1005Metric	117.1, 73.4	E 1090 接收链	
	C35	3pF	C_0402_1005Metric	115.1, 71.1	E 1090 接收链	
	C37	100nF	C_0402_1005Metric	113.9, 77.0	E 1090 接收链	
	C38	100pF	C_0402_1005Metric	111.6, 77.0	E 1090 接收链	
	C39	100pF	C_0402_1005Metric	118.2, 92.8	D 978 收发	
	C40	3.6pF	C_0402_1005Metric	110.7, 94.2	D 978 收发	
	C41	2.7pF	C_0402_1005Metric	115.9, 97.5	D 978 收发	
	C42	6.2pF	C_0402_1005Metric	115.9, 95.2	D 978 收发	
	C43	3pF	C_0402_1005Metric	115.9, 92.8	D 978 收发	
	C44	3.6pF	C_0402_1005Metric	112.8, 96.5	D 978 收发	
	C45	100pF	C_0402_1005Metric	110.7, 90.2	D 978 收发	
	C46	3pF	C_0603_1608Metric	104.9, 78.5	E 1090 接收链	板上印着 C47，尺寸相同会真贴错
	C47	200pF	C_0603_1608Metric	104.9, 80.4	E 1090 接收链	板上印着 C46，尺寸相同会真贴错
	C49	100nF	C_0603_1608Metric	77.0, 75.8	E 1090 接收链	
	C51	1nF	C_0603_1608Metric	77.0, 81.0	E 1090 接收链	
	C52	100nF	C_0402_1005Metric	99.6, 82.3	E 1090 接收链	

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	C53	100pF	C_0402_1005Metric	93.6, 62.9	E 1090 接收链	用J7测IFA时不贴，避免后级并入VNA读数
	C54	100pF	C_0402_1005Metric	103.9, 61.7	E 1090 接收链	
	C55	100pF	C_0603_1608Metric	97.6, 66.5	E 1090 接收链	
	C56	100pF	C_0603_1608Metric	95.0, 66.5	E 1090 接收链	
	C57	100pF	C_0402_1005Metric	68.3, 81.0	C 传感器 + GNSS	
	C58	100pF	C_0402_1005Metric	57.7, 76.5	C 传感器 + GNSS	
	C59	100pF	C_0402_1005Metric	57.6, 69.8	C 传感器 + GNSS	
	C60	22uF	C_0805_2012Metric	99.3, 98.0	D 978 收发	
	C61	100nF	C_0603_1608Metric	102.7, 99.1	D 978 收发	
	C62	100nF	C_0603_1608Metric	105.4, 98.3	D 978 收发	
	C63	100nF	C_0603_1608Metric	107.1, 84.6	D 978 收发	
	C64	100nF	C_0603_1608Metric	100.2, 85.8	D 978 收发	
	C65	100nF	C_0603_1608Metric	99.1, 94.7	D 978 收发	
	C66	22uF	C_0805_2012Metric	102.9, 97.2	D 978 收发	
	C67	1uF	C_0603_1608Metric	103.6, 85.8	D 978 收发	
	C68	18pF	C_0603_1608Metric	113.2, 85.0	D 978 收发	
	C69	18pF	C_0603_1608Metric	110.1, 85.0	D 978 收发	
	C70	47nF	C_0603_1608Metric	135.7, 80.2	A 电源	
	C71	4.7uF	C_0603_1608Metric	133.4, 79.0	A 电源	
	C72	6.8uF	C_1206_3216Metric	130.9, 79.2	A 电源	
	C73	10uF	C_0805_2012Metric	141.3, 88.1	A 电源	
	C74	10uF	C_0805_2012Metric	143.3, 84.9	A 电源	
	C75	10uF	C_0805_2012Metric	139.1, 88.1	A 电源	
	C76	22uF	C_1206_3216Metric	134.4, 96.3	A 电源	
	C77	22uF	C_1206_3216Metric	141.2, 99.7	A 电源	
	C78	1uF	C_0603_1608Metric	134.4, 94.2	A 电源	
	C79	1uF	C_0603_1608Metric	130.6, 65.0	A 电源	
	C80	1uF	C_0603_1608Metric	136.3, 74.6	A 电源	
	R1	10k	R_0603_1608Metric	55.2, 84.5	C 传感器 + GNSS	
	R2	4.7k	R_0603_1608Metric	64.1, 97.8	C 传感器 + GNSS	
	R3	4.7k	R_0603_1608Metric	64.1, 95.2	C 传感器 + GNSS	
	R4	1k	R_0603_1608Metric	82.0, 69.1	B MCU + Flash	
	R5	10k	R_0603_1608Metric	80.0, 73.2	B MCU + Flash	
	R6	10k	R_0603_1608Metric	87.9, 86.4	B MCU + Flash	
	R7	5.1k	R_0603_1608Metric	145.6, 64.8	B MCU + Flash	
	R8	5.1k	R_0603_1608Metric	143.6, 64.8	B MCU + Flash	
	R9	27R	R_0603_1608Metric	87.9, 83.3	B MCU + Flash	
	R10	27R	R_0603_1608Metric	85.3, 83.3	B MCU + Flash	
	R17	10k	R_0603_1608Metric	121.8, 98.7	D 978 收发	
	R18	10k	R_0603_1608Metric	91.4, 71.4	E 1090 接收链	
	R19	52.3R	R_0402_1005Metric	115.1, 73.4	E 1090 接收链	
	R21	0R	R_0603_1608Metric	105.0, 72.6	E 1090 接收链	
	R22	0R	R_0603_1608Metric	100.3, 66.5	E 1090 接收链	
	R23	0R	R_0603_1608Metric	93.2, 77.2	E 1090 接收链	
	R24	0R DNP(旁路外接)	R_0402_1005Metric	101.1, 63.1	不贴 (DNP)	
	R25	0R DNP(旁路板载)	R_0402_1005Metric	95.5, 62.9	不贴 (DNP)	
	R26	10k	R_0603_1608Metric	67.3, 73.7	C 传感器 + GNSS	
	R27	10k	R_0603_1608Metric	67.3, 71.1	C 传感器 + GNSS	
	R30	1k	R_0402_1005Metric	96.8, 77.2	E 1090 接收链	

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	R31	1k	R_0603_1608Metric	101.6, 75.7	E 1090 接收链	
	R32	10k	R_0603_1608Metric	104.9, 75.7	E 1090 接收链	
	R33	100k	R_0603_1608Metric	100.9, 82.7	E 1090 接收链	板上印着 C25, 尺寸相同会真贴错
	R34	10k	R_0603_1608Metric	77.0, 73.2	E 1090 接收链	
	R35	10k	R_0603_1608Metric	77.0, 78.4	E 1090 接收链	
	R36	1k	R_0603_1608Metric	96.5, 81.8	E 1090 接收链	
	R37	6.8k	R_0603_1608Metric	137.0, 66.4	A 电源	
	R38	180R	R_0603_1608Metric	133.1, 88.2	A 电源	
	R39	5.62k	R_0603_1608Metric	136.2, 88.2	A 电源	
	R40	31.6k	R_0603_1608Metric	134.6, 88.2	A 电源	
	R41	470k	R_0603_1608Metric	139.1, 98.6	A 电源	
	R42	150k	R_0603_1608Metric	137.4, 98.6	A 电源	
	R43	1k	R_0603_1608Metric	136.3, 72.0	A 电源	
	R44	10k	R_0603_1608Metric	137.0, 69.5	A 电源	
	R45	10k	R_0603_1608Metric	129.2, 84.0	A 电源	
	R46	10k	R_0603_1608Metric	129.2, 86.5	A 电源	
	L2	33nH	L_0402_1005Metric	57.7, 75.4	C 传感器 + GNSS	
	L7	6.8uH	L_0805_2012Metric	97.3, 94.7	D 978 收发	
	L8	100nH	L_0402_1005Metric	119.3, 90.3	D 978 收发	
	L9	7.5nH	L_0402_1005Metric	112.3, 94.8	D 978 收发	
	L10	27nH	L_0402_1005Metric	110.9, 95.7	D 978 收发	
	L11	6.8nH	L_0402_1005Metric	118.2, 97.5	D 978 收发	
	L12	6.8nH	L_0402_1005Metric	118.2, 95.2	D 978 收发	
	L13	7.5nH	L_0402_1005Metric	110.7, 92.2	D 978 收发	
	L14	100nH	L_0402_1005Metric	101.1, 70.6	E 1090 接收链	板上印着 C30, 尺寸相同会真贴错
	L15	33nH	L_0402_1005Metric	57.6, 68.7	C 传感器 + GNSS	
	L16	1uH	L_Bourns--SRN4018	139.7, 83.6	A 电源	
	L17	1.5uH	L_Bourns--SRN4018	144.8, 95.0	A 电源	
	D2	ESD 3.3V/0.6pF	D_0402_1005Metric	101.1, 72.6	E 1090 接收链	有极性, 认阴极标记带
	D3	ESD 3.3V/0.6pF	D_0402_1005Metric	117.5, 89.9	D 978 收发	有极性, 认阴极标记带
	F2	6V/200mA	Fuse_0805_2012Metric	121.9, 92.5	D 978 收发	
	F3	6V/200mA	Fuse_0805_2012Metric	98.3, 71.7	E 1090 接收链	
	F4	6V/200mA	Fuse_0805_2012Metric	60.3, 75.9	C 传感器 + GNSS	
	F5	6V/200mA	Fuse_0805_2012Metric	60.3, 69.2	C 传感器 + GNSS	
	Q2	AO3401A	SOT-23	121.9, 95.8	D 978 收发	
	Q3	AO3401A	SOT-23	95.3, 71.7	E 1090 接收链	
	Q4	AO3401A	SOT-23	63.5, 75.9	C 传感器 + GNSS	
	Q5	AO3401A	SOT-23	63.5, 69.2	C 传感器 + GNSS	
	U1	ME6211C33M5	SOT-23-5	74.7, 97.8	A 电源	认 pin1 (缺角/圆点)
	U2	TPS7A2033PDBVR	SOT-23-5	82.3, 97.8	A 电源	认 pin1 (缺角/圆点)
	U3	TPS7A2033PDBVR	SOT-23-5	89.2, 97.8	A 电源	认 pin1 (缺角/圆点)
	U4	BNO085	BNO085_LGA-28	59.7, 84.5	C 传感器 + GNSS	认 pin1 (缺角/圆点)
	U5	BMP388	BMP388_LGA-10	61.0, 88.9	C 传感器 + GNSS	认 pin1 (缺角/圆点)
	U6	QMC5883P	LGA-16_3x3mm_P0.5mm	59.7, 96.5	C 传感器 + GNSS	认 pin1 (缺角/圆点)
	U7	ATGM336H-6N-74	ATGM336H_LCC-18	73.7, 89.1	C 传感器 + GNSS	认 pin1 (缺角/圆点)
	U8	RP2040	QFN-56-1EP_7x7mm_P0.4mm_EP3.2	86.4, 76.9	B MCU + Flash	认 pin1 (缺角/圆点)
	U9	W25Q128JVS1Q	SOIC-8_5.3x5.3mm_P1.27mm	93.7, 85.9	B MCU + Flash	认 pin1 (缺角/圆点)

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	U10	CC1312R1F3RGZR	QFN-48-1EP_7x7mm_P0.5mm_EP5.1	104.5, 92.0	D 978 收发	认 pin1 (缺角/圆点)
	U11	QPL9547TR7	DFN-8-1EP_2x2mm_P0.5mm_EP0.8	107.7, 61.9	E 1090 接收链	认 pin1 (缺角/圆点)
	U12	BGA2817	SOT-363_SC-70-6	117.1, 62.5	E 1090 接收链	认 pin1 (缺角/圆点)
	U14	AD8313ARMZ	MSOP-8_3x3mm_P0.65mm	111.1, 73.4	E 1090 接收链	认 pin1 (缺角/圆点)
	U15	TLV3501(TOKMAS)	SOT-23-6	100.6, 79.5	E 1090 接收链	认 pin1 (缺角/圆点)
	U16	XA17-G4K(或AS179-92LF)	SOT-363_SC-70-6	98.0, 62.9	E 1090 接收链	认 pin1 (缺角/圆点)
	U17	XA17-G4K(或AS179-92LF)	SOT-363_SC-70-6	68.6, 78.1	C 传感器 + GNSS	认 pin1 (缺角/圆点)
	U18	CH224K	CH224K_ESSOP-10	132.5, 68.6	A 电源	认 pin1 (缺角/圆点)
	U19	SY6970	QFN-24-1EP_4x4mm_P0.5mm_EP2.6	134.5, 83.7	A 电源	认 pin1 (缺角/圆点)
	U20	SY7069	TSOT-23-6	139.2, 95.0	A 电源	认 pin1 (缺角/圆点)
	FL1	TA0970A	TA0970A_SMD3838-6	112.0, 65.5	E 1090 接收链	SAW 滤波器分输入输出, 贴反不工作
	FL2	TA0970A	TA0970A_SMD3838-6	118.4, 66.5	E 1090 接收链	SAW 滤波器分输入输出, 贴反不工作
	Y1	12MHz	Crystal_SMD_3225-4Pin_3.2x2.5mm	85.1, 69.1	B MCU + Flash	无源晶体, 转 180° 等价, 长边对长边即可
	Y2	48MHz ABM8W-7pF	Crystal_SMD_3225-4Pin_3.2x2.5mm	108.3, 98.3	D 978 收发	无源晶体, 转 180° 等价, 长边对长边即可
	Y3	32.768kHz FC-135	Crystal_SMD_3215-2Pin_3.2x1.5mm	111.6, 87.0	D 978 收发	
	J1	J3_HAT_2x20_SMD排针	PinHeader_2x20_P2.54mm	92.6, 106.3	F 对外接口	
	J2	U.FL_GNSS_EXT(经尾线转SMA)	U.FL_Hirose_U.FL-R-SMT-1_Vertical	54.0, 75.9	C 传感器 + GNSS	
	J4	USB-C_16P	USB_C_Receptacle_HRO_T C-31-M-12	145.8, 73.0	B MCU + Flash	
	J5	U.FL_978	U.FL_Hirose_U.FL-R-SMT-1_Vertical	118.5, 86.3	D 978 收发	
	J6	U.FL_1090_EXT(经尾线转SMA)	U.FL_Hirose_U.FL-R-SMT-1_Vertical	106.5, 67.3	E 1090 接收链	
	J7	U.FL_IFA_TEST(π后调试口)	U.FL_Hirose_U.FL-R-SMT-1_Vertical	90.0, 61.3	E 1090 接收链	
	J8	U.FL→内置patch	U.FL_Hirose_U.FL-R-SMT-1_Vertical	54.0, 69.2	C 传感器 + GNSS	
	J9	MX1.25WT-2P BAT	MX1.25WT-2P_1x02-1MP_P1.25mm_Horizontal	134.6, 108.7	A 电源	
	ANT1	IFA_1090	ANT_IFA_1090MHz	79.2, 57.8	不贴 (板载天线)	52.0mm画长/53.5mm外包络已落PCB; 待装盒 VNA+报文A/B, 不是已调准
	ZP1	DNP 并-天线侧	C_0603_1608Metric	79.2, 63.6	不贴 (DNP)	默认DNP; HFSS首轮仍DNP
	ZP2	DNP 并-电台侧	C_0603_1608Metric	84.2, 63.6	不贴 (DNP)	默认DNP; 完整六层rev2 HFSS首轮可从3.3pF起扫, 并备3.6/3.9pF
	ZS1	0R 串	L_0603_1608Metric	81.7, 62.9	E 1090 接收链	默认0R直通; HFSS首轮可从3.6nH起扫
	SW2	BOOTSEL短接焊盘	SolderJumper-2_P1.3mm_Open_Pad1.0x1.5	51.9, 81.0	不贴 (短接焊盘)	
	H1	—	MountingHole_2.7mm_M2.5	54.0, 56.0	不贴 (安装孔)	
	H2	—	MountingHole_2.7mm_M2.5	146.0, 56.0	不贴 (安装孔)	
	H3	—	MountingHole_2.7mm_M2.5	54.0, 106.0	不贴 (安装孔)	
	H4	—	MountingHole_2.7mm_M2.5	146.0, 106.0	不贴 (安装孔)	
	FID1	—		59.2, 56.0	??	
	FID2	—		140.8, 56.0	??	
	FID3	—		59.2, 106.0	??	
	RT1	NCP18XH103F03RB 10k NTC	R_0603_1608Metric	120.9, 82.0	A 电源	